

Exercise 4

1a) • The feasible set is $x \in [2, 4]$

• Optimal value and solution in the feasible set:

$$x = 2; f(x) = 2^2 + 1 = 5$$

$$\begin{aligned} \text{b) } L(x, \lambda) &= x^2 + 1 + \lambda(x-2)(x-4) = x^2 + 1 + \lambda(x^2 - 6x + 8) \\ &= x^2(1+\lambda) + 1 - 6\lambda x + 8\lambda \end{aligned}$$

$$g(\lambda) = \inf_{x \in D} L(x, \lambda)$$

$$= \inf_{x \in D} x^2 + 1 + \lambda(x^2 - 6x + 8)$$

$$\frac{\partial L}{\partial x} = 2x + \lambda(2x - 6) = 0$$

$$2x(1+\lambda) = 6\lambda$$

$$x = \frac{6\lambda}{2(1+\lambda)} = \frac{3\lambda}{1+\lambda}$$

→ assume $\lambda > -1$

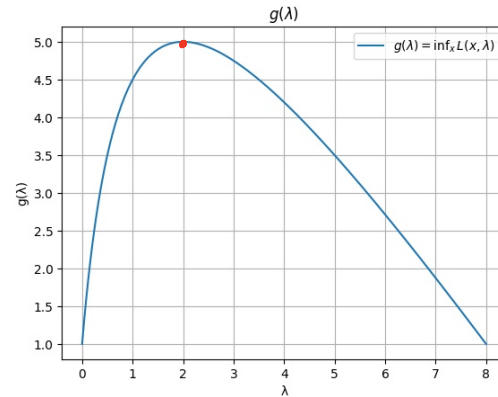
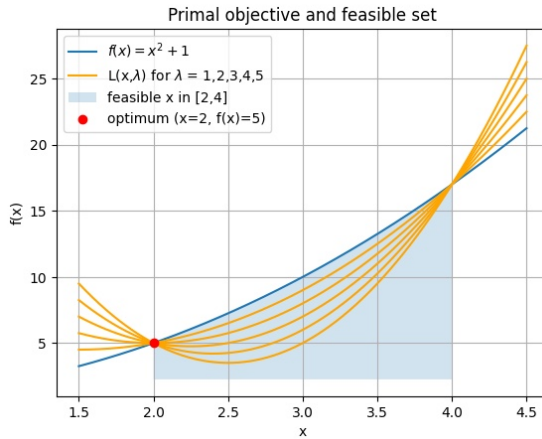
$$g(\lambda) = \left(\frac{3\lambda}{1+\lambda}\right)^2 (1+\lambda) + 1 - 6\lambda \frac{3\lambda}{1+\lambda} + 8\lambda$$

$$= \frac{9\lambda^2}{1+\lambda} + 1 - \frac{18\lambda^2}{1+\lambda} + 8\lambda$$

$$= -\frac{9\lambda^2}{1+\lambda} + 1 + 8\lambda$$

$$g(\lambda) = \begin{cases} -\frac{9\lambda^2}{1+\lambda} + 1 + 8\lambda & \text{for } \lambda > -1 \\ -\infty & \text{else} \end{cases}$$

plots



$$\rightarrow g(\lambda) \leq 5 = p^*$$

c) Dual problem:

maximize $g(\lambda)$ subject to $\lambda \geq 0$

$$g(\lambda) = -\frac{9\lambda^2}{1+\lambda} + 1 + 8\lambda$$

$$\frac{2g}{2\lambda} = -\frac{18\lambda(1+\lambda) - 9\lambda^2}{(1+\lambda)^2} + 8 = -\frac{18\lambda^2 + 18\lambda - 9\lambda^2}{(1+\lambda)^2} + 8$$

$$= -\frac{9\lambda(\lambda+2)}{(1+\lambda)^2} + 8$$

$$\frac{2g^2}{2\lambda} = -\frac{18}{(1+\lambda)^3} \leq 0 \quad \text{for } \lambda \geq 0 \Rightarrow \text{is concave}$$

Optimal value / solution:

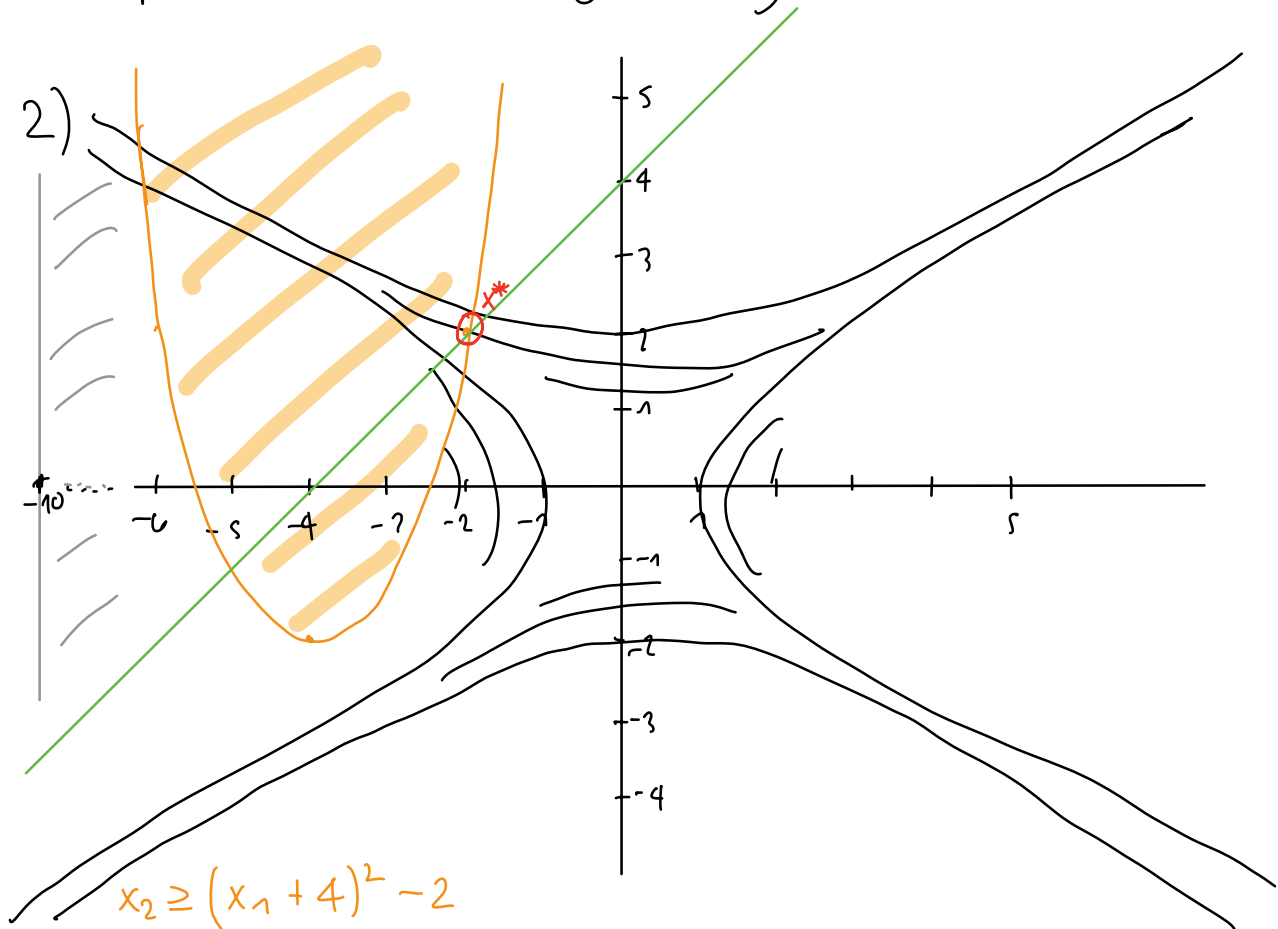
$$g = \frac{9\lambda(\lambda+2)}{(1+\lambda)^2} \quad \Leftrightarrow 8\lambda^2 + 16\lambda + 8 = 9\lambda^2 + 18\lambda$$

$$\Rightarrow 8 = \lambda^2 + 2\lambda$$

$$8 = \lambda(\lambda+2) \quad \Rightarrow \quad \lambda = 2$$

$$g(2) = 5 = d^*$$

$\Rightarrow p^* = d^* \Rightarrow$ strong duality



$$2) \quad x^* = (-2, 2)$$

$$(1) \text{ primal constraints: } \begin{aligned} (x_1 + 4)^2 + 2 - x_2 &\leq 0 & f_1 \\ -10 - x_1 &\leq 0 & f_2 \\ x_1 - x_2 + 4 &= 0 & h_1 \end{aligned}$$

$$(2) \text{ dual constraints: } \lambda_1 \geq 0, \lambda_2 \geq 0$$

$$(3) \text{ complementary slackness: } \lambda_1 \underbrace{(-2 - (-2 + 4)^2 - 2)}_{=0} = 0 \Rightarrow \lambda_1 \text{ free}$$

$$\lambda_2 (-10 - 2) = 0$$

$$\Rightarrow \lambda_2 = 0$$

(4) vanishing gradient of Lagrangian

$$L(x, \lambda, v) = x_1^2 - 2x_2^2 + \lambda_1 ((x_1 + 4)^2 + 2 - x_2) + \lambda_2 (-10 - x_1) + v (x_1 - x_2 + 4)$$

$$\nabla L = \begin{pmatrix} 2x_1 \\ -4x_2 \end{pmatrix} + \lambda_1 \begin{pmatrix} 2x_1 + 8 \\ -1 \end{pmatrix} + \lambda_2 \begin{pmatrix} -1 \\ 0 \end{pmatrix} + v \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Find dual variables:

$$\underline{\lambda_2 = 0} \quad (2) \quad h_1: x_1 = x_2 - 4$$

$$\begin{aligned} \text{I} \quad & \left\| \begin{aligned} 2x_2 - 8 + \lambda_1(2x_2 - 8) + 8 + v &= 0 \\ -4x_2 - \lambda_1 - v &= 0 \end{aligned} \right\| \Rightarrow v = -8 - \lambda_1 \\ \text{II} \quad & \begin{aligned} -4 + 4\lambda_1 + 8 + v &= 0 \\ 4\lambda_1 - 4 - 8 - \lambda_1 &= 0 \Rightarrow \underline{\lambda_1 = 4} \\ \Rightarrow v = -8 - 4 &= \underline{\underline{-12}} \end{aligned} \end{aligned}$$